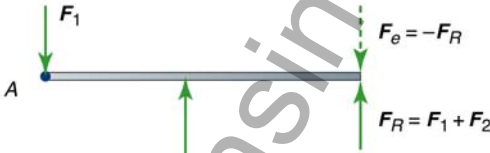


Equilibrium



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ME101 (Spring 2017)

Basic Mech. Engg.

Equilibrium

EQUATIONS

$$\sum \vec{F} = m\vec{a}$$

$$\sum \vec{M} = I\vec{\alpha}$$

In Equilibrium $a = 0$ and $\alpha = 0$

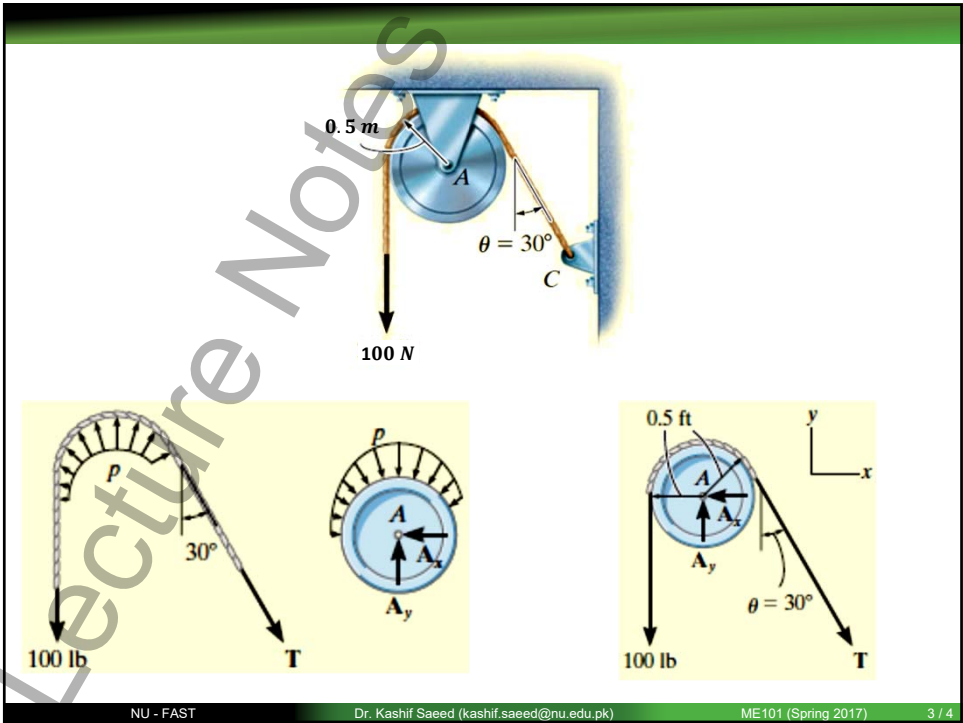
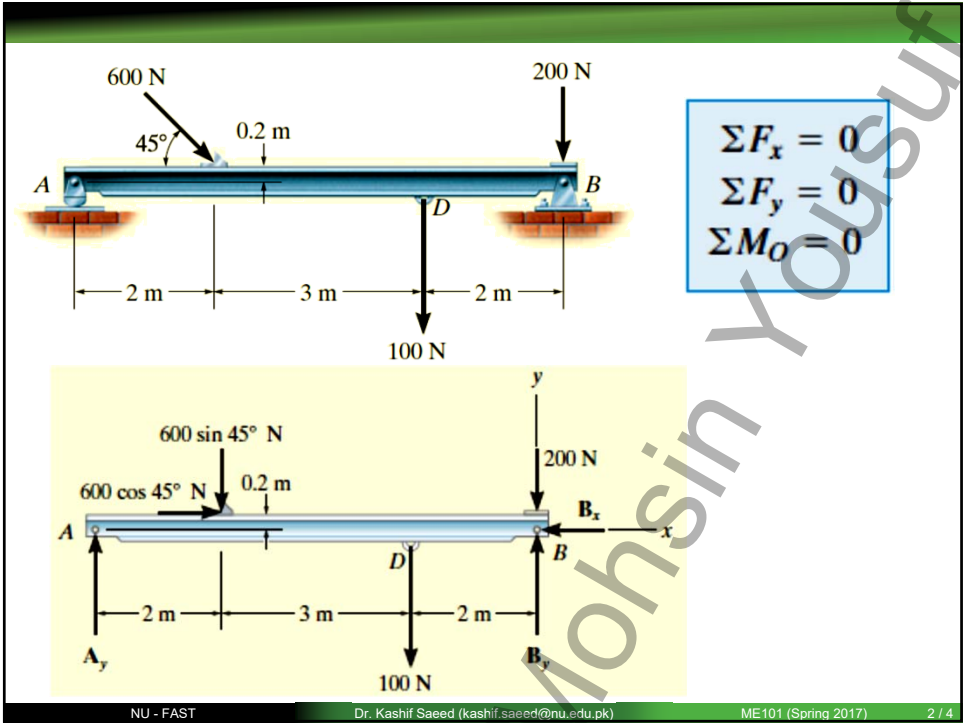
$$\sum \vec{F} = 0$$

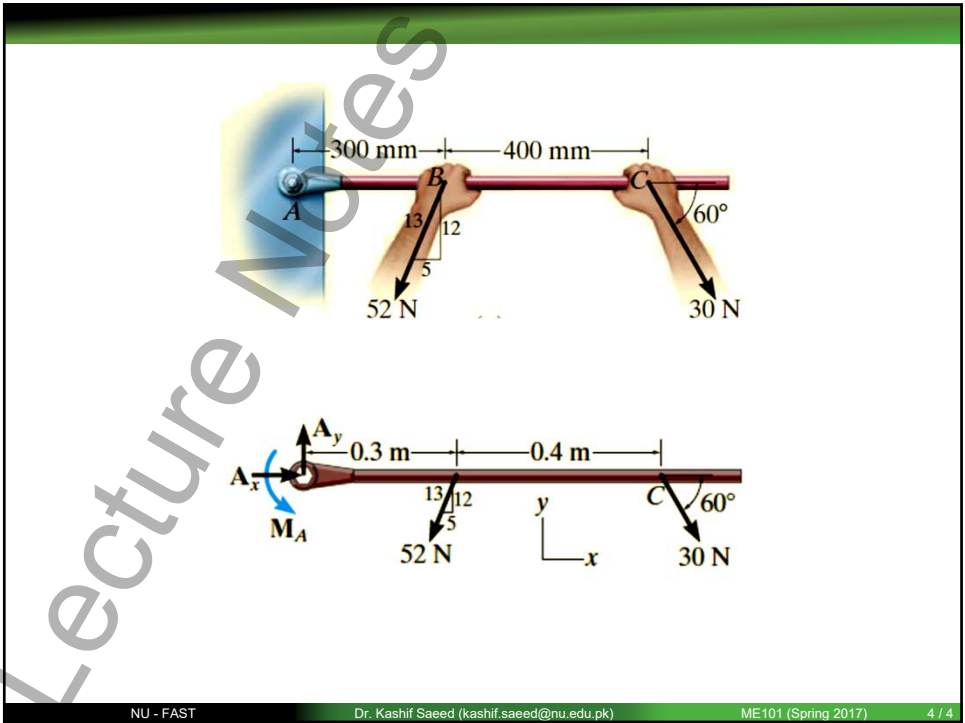
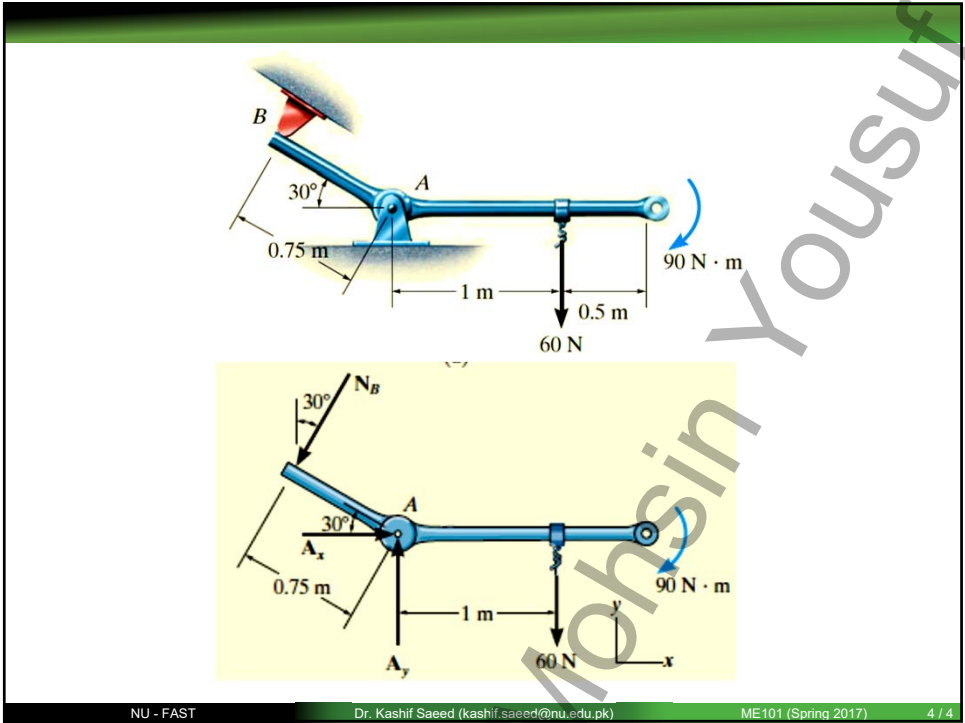
$$\sum \vec{M} = 0$$

How many equations are there?

six

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Q. The rod is supported by a smooth wall and at B and C either at the top or bottom by rollers, determine the reactions at these supports. Neglect the weight of the rod.

Diagram showing a rod supported by a smooth wall and rollers. The rod is inclined at 30° . A 300 N force acts vertically downwards at point C . A $4000\text{ N}\cdot\text{m}$ clockwise moment acts at point B . The rod is supported by a smooth wall at point A and rollers at points B and C . Dimensions are given in meters: 2 m from C to B , 4 m from B to A , and 2 m from A to the wall. A free-body diagram is shown to the right with reaction forces A_x , A_y , B_x , B_y , and C_x , C_y .

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Q. The collar at A is fixed to the member and can slide vertically along the vertical shaft, determine the support reactions on the member.

Diagram showing a frame structure. A horizontal member of length 3 m is supported by a collar at point A that can slide vertically along a vertical shaft. A 900 N force acts vertically downwards at the midpoint (1.5 m from A). A $500\text{ N}\cdot\text{m}$ clockwise moment acts at the midpoint. The member is connected to a vertical member of length 1 m at point B , which is inclined at 45° . A free-body diagram is shown below with reaction forces A_x , A_y , and N_B .

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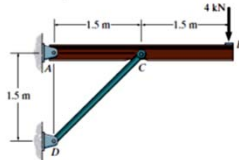
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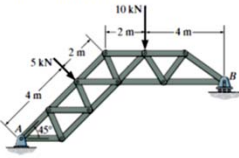
PRACTICE PROBLEMS

Problems 5.2 to 5.6

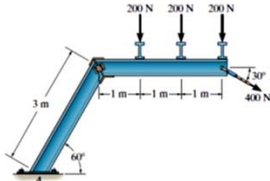
F5-2. Determine the horizontal and vertical components of reaction at the pin A and the reaction on the beam at C.



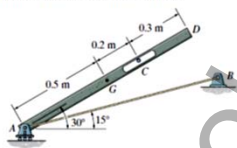
F5-3. The truss is supported by a pin at A and a roller at B. Determine the support reactions.



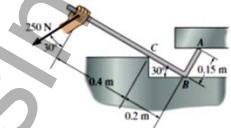
F5-4. Determine the components of reaction at the fixed support A. Neglect the thickness of the beam.



F5-5. The 25-kg bar has a center of mass at G. If it is supported by a smooth peg at C, a roller at A, and cord AB, determine the reactions at these supports.



F5-6. Determine the reactions at the smooth contact points A, B, and C on the bar.



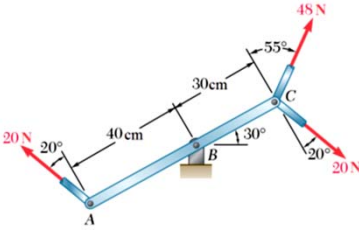
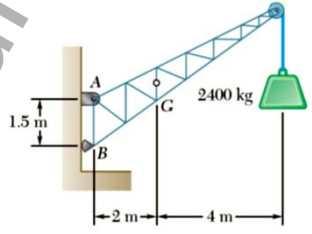
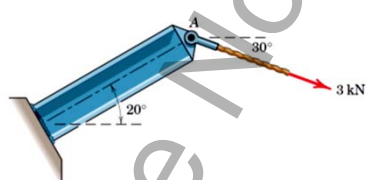
Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp 226

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MORE PRACTICE PROBLEMS



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Mohsin Yousuf

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